Question	Answer	Marks	Guidance
7(a)	current for which the direction regularly reverses	B1	**Allow** 'direction changes' for 'direction reverses' **Allow** 'periodically', 'repeatedly' or 'continuously' for 'regularly'
7(b)(i)	straight parallel horizontal lines shown inside the solenoid	B1	Minimum 5 lines needed inside. **Allow** curvature to start just before the ends.
	correct bar magnet field pattern shown outside the solenoid	B1	Minimum 5 lines needed outside. **Do not allow** lines merging or crossing At least one complete loop above and one below the central axis and one horizonal line outside each pole required. To award both of the B1 marks, the lines inside and outside must be continuous with each other.
	arrows showing direction inside the solenoid is right to left	B1	Arrows can be anywhere on field lines (whether inside or outside the solenoid) but must be consistent with correct direction (extrapolated if necessary) inside. Minimum of 1 arrow inside and 1 arrow outside needed, but all arrows shown must be in correct direction.
7(b)(ii)	N shown at left-hand end of solenoid and S shown at right-hand end	B1	No ECF from **7(b)(i)**. **Allow** omission of either N or S if the other letter is present in the correct place.
7(c)(i)	$200\pi = 2\pi / T$ so $T = 0.010$ s	A1	Full substitution and answer needed; π must be shown on both sides of the equation. **Allow** 0.01 s

Question	Answer	Marks	Guidance
7(c)(ii)	two cycles of a sinusoidal curve drawn, with period 0.010 s and amplitude 2.0 mT	B1	**Allow** tolerance ±½ small square. Judge period on the crossing points of the *t*-axis. 'Sinusoidal' judged by curvature at peaks and troughs which are not at edges of the grid.
	correct phase, with $B = 0$ at (and only at) $t = 0.0025$ s, $t = 0.0075$ s, $t = 0.0125$ s and $t = 0.0175$ s.	B1	**Ignore** amplitude and times of peaks / troughs. Assess phase only by the crossing points of *t*-axis.
7(d)(i)	$\Phi = NBA$	C1	**Allow** $\Phi = BA$ without N for the first C1 mark
	$= 45 \times 2.0 \times 10^{-3} \times \pi \times (2.8 \times 10^{-3})^2$	C1	**Ignore** POT in maximum flux density in second C1 mark
	$= 2.2 \times 10^{-6}$ Wb	A1	Correct to at least 2 significant figures. AFC. (2.22) Unit needed with answer for the A1 mark. **Allow** 'T m^2' or 'V s' or 'kg m^2 s^{-1}' for 'Wb'. **Allow** WB for Wb The POT in the answer must be consistent with the unit given, e.g. 2.2×10^{-3} mWb
7(d)(ii)	draw tangent to determine gradient at steepest part of curve	B1	
	steepest gradient × area (of search coil) × number (of turns) gives maximum rate of cutting flux	B1	**Accept** '*A*' for 'area' and '*N*' for 'number'
	maximum e.m.f. = maximum rate of cutting flux (by Faraday's law)	B1	**Accept** symbol for maximum e.m.f. <u>Alternative method for the 2nd and 3rd B1 marks:</u> e.m.f = rate of cutting flux (B1) (**Do not allow** in symbols because Φ defined in question as maximum flux) maximum e.m.f = steepest gradient × area × number (B1) (**Accept** symbol for maximum e.m.f., '*A*' for 'area' and '*N*' for 'number') Both marks must be from the same method.